

CFD 2025-26 – Lab 04

Exercise 1

Let us consider the Stokes problem defined in the unit square $\Omega := (0, 1)^2$

$$\begin{cases} -\Delta \mathbf{u} + \nabla p = 0, & \text{in } \Omega, \\ \operatorname{div} \mathbf{u} = 0, & \text{in } \Omega, \\ \mathbf{u} = 1\mathbf{i} + 0\mathbf{j}, & \text{on } \Gamma^{\text{up}} = \{0 \leq x \leq 1, y = 1\}, \\ \mathbf{u} = \mathbf{0}, & \text{on } \partial\Omega \setminus \Gamma^{\text{up}}. \end{cases}$$

- 1 Assemble the monolithic block matrix $\Sigma = \begin{bmatrix} A & B^T \\ -B & 0 \end{bmatrix}$ associated to the problem using the stable pair of spaces $\mathbb{P}^2/\mathbb{P}^1$. Solve the system using the GMRES method.
- 2 Using a value of 10^{-8} for the tolerance, does the number of iterations needed to reach the convergence increase with the number of the elements? (Use a uniformly refined grid with $n = 10, 20, 40, 80$ subdivisions). Use the **Firedrake** function `LinearVariationalSolver` with the following parameters:

```
{'ksp_type': 'gmres', 'pc_type': 'none',  
 'ksp_rtol': 1.e-8, 'ksp_max_it': 10000}
```

- 3 Now let us introduce the block diagonal preconditioner

$$P = \begin{bmatrix} A & 0 \\ 0 & \frac{1}{\nu} M_p \end{bmatrix},$$

where M_p is the mass matrix for the pressure unknown and ν is the kinematic viscosity of the problem. Does the number of iterations needed to reach the convergence increase with the number of the elements? (Use a uniformly refined grid with $n = 10, 20, 40, 80$ subdivisions and for the tolerance consider a value of 10^{-8}).

- 4 Study the eigenvalues for both the unpreconditioned and preconditioned system.
- 5 Modify the code to implement the block *triangular* preconditioner

$$P = \begin{bmatrix} A & 0 \\ -B & \frac{1}{\nu} M_p \end{bmatrix}.$$

In the parameter `pc_fieldsplit_type`, the value `multiplicative` has to be used, instead of `additive`. What can you observe?